From Reasoning with Cases to Retrieval-Augmented Law: A Survey of the Evolution of Legal Artificial Intelligence (1991–2026)

Ali Hamza

Department of Software Engineering, Military College of Signals (MCS),

National University of Sciences and Technology (NUST), Islamabad, Pakistan

ahamza.msse25mcs@student.nust.edu.pk

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Abstract

Legal work depends on finding, interpreting, and applying authoritative texts. Lawyers and judges work with statutes, regulations, prior decisions, and procedural rules. A useful AI system must therefore do more than produce fluent language: it must connect its claims to relevant and reliable legal sources. This survey traces the development of legal artificial intelligence from case-based reasoning in the early 1990s to contemporary legal language models and retrieval-augmented generation (RAG). Rather than presenting a simple chronology, we explain how each wave of methods responded to limitations in earlier approaches. We review foundational reasoning systems, legal language models, benchmarks, legal information retrieval, RAG, and studies of hallucination and citation reliability. We then examine three recent system designs that improve different parts of the pipeline: document drafting with RAG, fine-tuning of retrieval and generation, and structure-aware indexing. The review highlights persistent challenges in statutory hierarchy preservation, citation verification, multilingual access, and jurisdictional coverage, with particular attention to under-represented legal systems such as Pakistan. We distinguish findings reported by the original authors from limitations inferred from a study's scope or evaluation.

1 Introduction

Law is a language-intensive domain. Statutes define legal duties, judgments explain how rules have been applied, and legal arguments connect facts to authoritative sources. These features made law an early candidate for artificial intelligence (AI). A computer system that can represent legal rules and compare cases could help practitioners locate relevant provisions, identify similar precedents, and prepare or review legal documents.

Yet legal work places unusually high demands on correctness. A fluent answer that cites the wrong article, overlooks an exception, or invents a judgment can mislead its reader. Legal AI must therefore satisfy two connected requirements: it should communicate clearly, and its claims should be traceable to sources that actually support them. The history of legal AI can be read as a continuing effort to meet both requirements.

Early systems represented legal reasoning explicitly, often by comparing a new dispute with earlier cases or applying encoded rules (Ashley, 1991, 1992). Later, machine-learning methods and legal language models made it possible to process larger collections of text (Chalkidis et al., 2017, 2020). However, a model's ability to produce plausible language does not guarantee that its answer is legally correct. Retrieval-augmented generation (RAG) addresses part of this problem by retrieving external material at answer time and supplying it to the generator (Lewis et al., 2020). Retrieval helps expose evidence, but it does not guarantee that the evidence is relevant, current, correctly interpreted, or cited accurately. This survey follows that progression and makes four contributions: it organizes work by methodological change; distinguishes models, benchmarks, retrieval systems, and reliability studies; compares three recent legal-system designs; and identifies research questions around structure, provenance, language, and jurisdiction.

The paper is organized as follows. Section 2 describes the scope and source-selection process. Section 3 reviews early case-based and rule-based systems. Section 4 covers legal language models and benchmarks. Section 5 discusses legal retrieval and RAG. Section 6 examines hallucination, relevance, and citation reliability. Section 7 considers statutory structure and jurisdiction and compares three recent system designs. Section 8 presents open research directions, and Section 9 concludes.

2 How we selected and verified the literature

A survey is only as reliable as its source selection. This is especially important in legal AI, where a central concern is whether a system's claims can be checked against authoritative evidence. We therefore treated

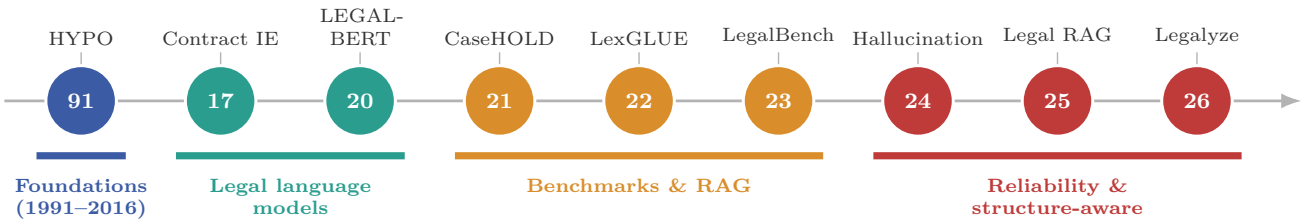


Figure 1: The evolution of legal AI, 1991–2026, read as a sequence of methods rather than dates. Four waves build on one another: transparent but non-scaling reasoning systems (*Foundations*); *legal language models* that scaled but were hard to trust; shared *benchmarks and retrieval-augmented generation*; and recent work on *reliability and structure-aware* retrieval. Only verified sources are placed on the line.

the literature as a set of records to identify, screen, and verify, rather than as a collection of titles assembled without a consistent method.

Search scope. The review covers work published from 1991 through October 2026. We considered foundational AI-and-law publications, legal NLP papers, legal information-retrieval research, RAG systems, benchmark and shared-task papers, and studies of reliability or hallucination. Sources included the ACL Anthology, publisher platforms, AI-and-law conference proceedings, COLIEE materials, and relevant preprint repositories.

Eligibility. A paper was relevant if it contributed to at least one of five themes: (1) legal reasoning and expert systems; (2) legal language models; (3) legal information retrieval or RAG; (4) legal benchmarks and evaluation; or (5) reliability, citation correctness, or governance. We gave particular attention to work that addresses statutory structure, evidence provenance, multilingual use, and legal systems beyond the most frequently studied English-language jurisdictions.

Record checking and deduplication. For each included work, we checked the title, author information, publication venue, year, and DOI or stable URL against the publisher, conference proceedings, or repository record. Duplicate records were removed using DOI and title. Where a detail could not be confirmed, it was not used as a factual claim. This is a structured narrative review, not a claim of exhaustive systematic coverage: database indexing, access restrictions, and the rapid pace of 2026 publications may mean that relevant work has been missed.

How limitations are reported. We separate limitations explicitly acknowledged by an article's authors from limitations inferred from its design or evaluation. For example, a benchmark restricted to one jurisdiction may have limited cross-jurisdictional generalizability; this is an analytical observation unless the authors state it themselves. This distinction helps keep comparisons fair and prevents an inference from being misrepresented as a reported result.

2.1 Related surveys and how this one differs

Several recent surveys map important parts of legal AI. Chi et al. (2026) organize research on legal LLMs around data, modeling, and evaluation. Zhao et al. (2026) review AI applications in litigation. Feng et al. (2024) and Yang et al. (2026) focus on legal case retrieval, while Premasiri et al. (2025) review legal information extraction. Kelsall et al. (2025) examine evaluation methods for LLMs in legal use cases. Together, these reviews show that the field is broad and that retrieval, generation, and evaluation are often studied in separate strands.

This survey connects those strands through one central question: how have legal AI methods tried to make answers both useful and traceable? It follows the progression from explicit reasoning, to learned legal representations, to retrieval-grounded generation and reliability assessment. It also compares three recent systems at the architectural level. The purpose is not to claim that one system is universally superior, but to clarify which design choices address generation quality, model adaptation, or preservation of legal structure.

Two commitments set the paper apart. The first is the verified-source discipline described above: every cited record was checked against its original. The second is a focus on the two gaps that the broad surveys name but do not dwell on, namely preserving the structure of statutes and serving jurisdictions outside the usual English-language data. Table 1 places the recent surveys side by side on these points.

3 Foundations: reasoning with cases and rules (1991–2016)

Early legal AI systems attempted to represent parts of legal reasoning directly. In common-law settings, arguments often depend on comparing a current dispute with earlier decisions: similarities may support an analogy, while differences may provide grounds for distinguishing a precedent. A computational system must therefore represent more than surface wording; it must capture facts, legally relevant dimensions, and the role those dimensions play in an argument.

The clearest example is the HYPO system described by Ashley (1991). HYPO modelled legal argument as a structured comparison between a current case and the cases already decided. It represented each case by a set of "dimensions" (factors that make a side stronger or weaker) and then built arguments by pointing to the most similar precedents and by raising hypothetical variations that tested how far a rule should stretch. The important idea here is that the system did not simply predict a winner; it produced an argument, with

Table 1: Recent surveys of legal AI and how this survey is positioned among them.

Survey	Scope	Main organizing idea	Structure / jurisdiction focus
Chi et al. (2026)	LLMs across legal data, modeling, and evaluation	Roles of LLMs: data, model, evaluator	Broad coverage; not focused on statutory structure
Zhao et al. (2026)	AI in litigation; systematic review	Five application areas of litigation AI	Application-led; jurisdiction coverage varies
Feng et al. (2024); Yang et al. (2026)	Legal case retrieval	Tasks, datasets, models, and metrics	Primarily case retrieval; cross-jurisdiction transfer remains a question
Premasiri et al. (2025)	Legal information extraction	Extraction tasks and open challenges	Identifies structure as a relevant processing challenge
Kelsall et al. (2025)	Evaluation of legal LLMs	Task definitions and evaluation metrics	Evaluation-led; practice relevance
This survey	Evolution toward fluent, traceable legal AI	Methodological waves plus a three-system comparison	Focus on structure preservation, provenance, and under-served jurisdictions

reasons, that a lawyer could read and challenge. Ashley (1992) later drew out what this means for legal expert systems more generally, explaining how a machine can assess relevance between cases and why case comparison, rather than fixed rules alone, fits the open texture of law.

Alongside case-based reasoning, the same period produced rule-based expert systems that encoded legal rules as explicit if-then logic. These systems were transparent, because every conclusion could be traced to a rule, but they were also brittle: real law is full of exceptions, vague terms, and conflicts between rules, and writing all of this by hand did not scale. As more legal text became available in digital form, attention moved from hand-written rules toward methods that could learn patterns from data.

Two lessons from this period remain relevant. First, a legal answer should be accompanied by an explanation that can be examined, not merely a conclusion. Second, legal reasoning depends on context and structure. Later neural systems can process far more text, but they do not automatically retain the explicit links between a legal claim, the rule it relies on, and the authority that establishes that rule.

4 Legal language models and the benchmarks that test them (2017–2023)

As machine learning became central to natural-language processing, legal researchers increasingly trained models to classify, extract, and represent information from legal documents. This shifted the emphasis from manually encoding every rule to learning patterns from examples. Tasks such as contract-element extraction showed that legal text could be processed with supervised models, although performance depended on the quality and scope of the annotated data (Chalkidis et al., 2017).

Pretrained language models extended this approach. General-purpose models learn broad language patterns, but legal documents contain specialized terms, long dependencies, and jurisdiction-specific conventions. LEGAL-BERT explored domain-specific pretraining and adaptation for legal language understanding (Chalkidis et al., 2020). Such models can improve downstream legal NLP tasks, but they do not by themselves provide an authoritative source for each generated claim. Learned legal knowledge and verifiable legal evidence are different things.

Benchmarks then became essential because results from separate datasets and tasks are difficult to compare. LexGLUE brought together multiple English-language legal understanding tasks under a shared evaluation framework (Chalkidis et al., 2022). LegalBench broadened the evaluation of LLMs across a diverse set of legal-reasoning tasks (Guha et al., 2023), while CaseHOLD examined whether models could identify the appropriate holding in a legal opinion (Zheng et al., 2021). These resources help researchers measure task performance, but no single benchmark captures every requirement of professional legal work, including source authority, citation support, jurisdictional fit, and temporal validity.

A parallel line of work came from the COLIEE competition, which runs each year on legal information extraction and entailment: retrieving relevant cases and statutes and deciding whether one text follows from another. The yearly overviews (Rabelo et al., 2022; Goebel et al., 2023, 2024, 2026) are useful because they show what methods win over time, from feature-based systems to transformer models and, more recently, to large language models. Table 2 summarises the main resources.

Another line of research represents legal concepts and their relationships explicitly. Knowledge graphs can link concepts, rules, and other entities, while structured retrieval can help answer questions that require relations across documents (Sovrano et al., 2020; Zhu et al., 2022). These approaches offer more explicit structure than a

plain text collection, but a graph of legal concepts and a hierarchy-preserving index solve different problems: the former emphasizes relationships, while the latter can retain the exact position of a provision within its source document.

5 Retrieval and retrieval-augmented generation for law (2020–2026)

Language models can produce plausible answers from patterns learned during training, but a user may need the rule that applies to a specific legal question today. RAG addresses this problem by retrieving external documents at inference time and providing them to the generator as context (Lewis et al., 2020). In legal applications, this can make answers easier to inspect and can help accommodate changes in statutes or case law without retraining the generator every time the corpus changes.

Retrieval is not a single method. Keyword search is useful when a query includes an exact section number, party name, or legal phrase. Dense retrieval can identify passages that are semantically related even when they use different wording. Hybrid retrieval combines these strengths; reranking and query rewriting can further improve the order or coverage of candidate passages. These methods should be evaluated separately from generation because a model cannot reliably use evidence that the retriever failed to find.

Recent legal RAG studies increasingly ask whether ordinary information-retrieval metrics reflect legal usefulness. Wiggers and Hartz (2024) examine relevance assessment for legal RAG, emphasizing that a passage can be topically similar without being legally useful. LR2 explores query rewriting and generated legal problems to support case-based reasoning (Benedetto et al., 2025). Buffa et al. (2025) apply retrieval and generation to judicial-document drafting. These systems differ in task and architecture, so their reported scores should not be compared directly unless datasets, queries, and evaluation protocols are aligned.

A related challenge is iterative retrieval. Zhang et al. (2025) explore a multi-round RAG approach for comprehensive legal-document analysis. Iteration may recover evidence missed in the first pass, but it can also add latency, complexity, and opportunities for error propagation. The practical question is not whether more retrieval is always better; it is whether the extra evidence improves legal correctness enough to justify its cost.

Jurisdiction is another major boundary. Legal systems differ in terminology, procedure, authority, and the relationship between statutes and precedent. A system developed for one country should not be assumed to work equally well in another. Research on Pakistani legal QA (Ali et al., 2025) is therefore important as a regional example, but its results must be judged by its corpus, evaluation design, and source-verification process rather than by the presence of RAG alone.

6 Reliability: hallucination and citation correctness

The strongest reason to retrieve legal sources is to make answers easier to check. Yet retrieval alone does not ensure that a generated statement is true. Dahl et al. (2024) document legal hallucinations in large language models, including unsupported or fabricated claims. Their findings demonstrate a reliability problem in the tested settings; they should not be treated as one universal hallucination rate for every model, jurisdiction, or legal task.

A natural next question is whether tools built specifically for legal research solve this problem. Magesh et al. (2025) evaluated leading commercial legal research products and found that their outputs were not uniformly reliable. The study is important because it evaluates deployed tools rather than only general-purpose chat models. Its findings are still tied to the products, questions, and scoring protocol used in the experiment, so the reported rates should be interpreted within that scope.

Reliability has several distinct dimensions. A source may exist but fail to support the claim; it may support the claim but be outdated; or it may be authoritative in one jurisdiction but irrelevant in another. Legal RAG evaluation should therefore separate retrieval relevance, answer correctness, citation existence, citation entailment, temporal validity, and jurisdictional validity. A single semantic-similarity score cannot represent all of these properties (Wiggers and Hartz, 2024).

Silva and Sanfilippo (2026) examine legal RAG in Brazilian civil procedure and show that semantic similarity and legal correctness can diverge. This result illustrates why evaluation should not rely on text overlap alone: an answer can sound similar to a reference answer and still misstate the applicable rule. The evaluation design and legal context matter when interpreting the result, and replication in other jurisdictions is needed before drawing broad conclusions.

Afane et al. (2026) evaluate AI systems on statutory-survey tasks in United States law and identify failures associated with both retrieval and reasoning. This distinction matters for system design. If the required provision is missing from the retrieved context, improving the generator alone may not solve the problem. If the evidence is present but misinterpreted, retrieval improvements alone may be insufficient. Reliable systems need separate tests for both stages and an end-to-end assessment of the final legal answer.

7 Structure, jurisdiction, and three design philosophies

The literature points to three connected challenges. First, legal documents have internal structure that can be lost during preprocessing and chunking. Second, most available datasets and evaluations cover only a subset of languages and jurisdictions. Third, a generated citation is not necessarily a verified citation. These issues are related: preserving a provision's location

Table 2: Representative legal benchmarks and shared tasks discussed in this survey.

Resource	Year	What it measures	Main caution
CaseHOLD (Zheng et al.)	2021	Selects the correct holding for a legal opinion	Multiple-choice holding prediction is narrower than full legal advice
COLIEE	Annual shared task	Legal case/statute retrieval and entailment	Task-specific datasets; results depend on competition track
LexGLUE	2022	Suite of English legal language-understanding tasks	Not designed to measure citation entailment or end-to-end RAG reliability
LegalBench	2023	A broad set of tasks for evaluating legal reasoning in LLMs	Task performance does not guarantee reliable real-world legal work
LeReRAG	2024	Legal relevance in retrieval-augmented generation	Evaluation scope and relevance labels need to be interpreted in context
Legal RAG Bench	2026	End-to-end legal RAG using a defined legal resource and complex questions	One resource/jurisdiction limits generalization

helps users inspect its context, while explicit provenance makes citation checking and evaluation more precise.

A statute is not simply a collection of independent paragraphs. Its parts, chapters, sections or articles, subsections, exceptions, definitions, and amendment history can affect interpretation. If a retrieval system separates a clause from its heading or parent provision, the retrieved text may lose context that changes its meaning. Structure-aware retrieval therefore aims to preserve both the text and its position in the legal document.

Different representations preserve different information. A knowledge graph emphasizes entities and relationships; a document tree captures parent-child structure; and a breadcrumb records the path from a provision to its enclosing parts and chapters. These approaches can complement one another, but they should not be treated as interchangeable. The appropriate representation depends on the query and on the legal relationships that must remain visible.

Legal language and jurisdiction also shape retrieval quality. A query written in English, Urdu, or Roman Urdu may refer to the same concept using different scripts or terminology. A model trained primarily on English-language law may fail to map those expressions to the correct local authority. This creates a need for language alignment, jurisdiction-aware metadata, and evaluations that test queries across languages rather than assuming that translation alone solves the problem.

Provenance should be preserved throughout the pipeline. A useful legal answer should identify the source document, exact provision or paragraph, jurisdiction, and, where relevant, the date or version of the law. A separate verification stage can then check that the source exists and supports the associated claim. This is stronger than asking a generator to produce citations as part of its prose and trusting those citations without checking them.

7.1 Three ways to build a modern legal system: Buffa et al., Ren et al., and Hamza et al.

These challenges reveal three design choices in recent legal systems. Buffa et al. (2025) use RAG to support legal-document drafting. Ren et al. (2026) adapt both retrieval and generation through fine-tuning with synthetic legal data. Hamza et al. (2026) focus on the index, preserving legal hierarchy and combining lexical and dense retrieval. The systems address different tasks and were not evaluated head-to-head on a common dataset; the comparison below is therefore architectural, not a ranking of performance.

Buffa et al.: improve generation at drafting time. Buffa et al. (2025) apply RAG to the construction of legal documents. The retrieval layer finds relevant material from precedent documents, while the RAG layer uses that material and the current document context to generate a draft section. This design can be updated by changing the source collection without retraining the language model. However, the quality of the draft still depends on what is retrieved and how faithfully the generator uses it. Users must check the resulting text against the relevant authority.

Ren et al.: adapt the model. Ren et al. (2026) combine synthetic legal QA generation and quality ranking with fine-tuning of retrieval and generation components. Their evaluation on Chinese legal benchmarks reports improvements from combining RAG with fine-tuning. This approach can make a model more suitable for a defined legal task, but it requires training data and computational resources. It also does not remove the need for evidence at inference time: fine-tuning may improve behavior, but it cannot guarantee that every answer or citation is correct, current, or supported.

Hamza et al.: adapt the index. Legalyze uses structure-aware hierarchical indexing (SAHI) to retain the position of a legal passage within its source. Each chunk carries metadata describing its place in the document, alongside a semantic anchor. The system combines BM25 keyword retrieval with dense vector retrieval and merges their rankings using reciprocal rank fusion. This design aims to improve access to exact legal references while retaining semantic matching. Its main dependency is the quality of parsing and metadata as-

signment: incorrect hierarchy information can mislead retrieval, so parser accuracy and citation verification remain important evaluation targets.

7.2 What the comparison tells us

Table 3 compares the systems across design dimensions that matter for legal applications. Because the studies use different tasks, corpora, and evaluation procedures, the table compares architecture and reported scope rather than claiming that one system achieves better overall performance.

Buffa et al. and Ren et al. focus primarily on the generation/model side, although they do so differently. Buffa et al. use retrieval context to guide drafting without fine-tuning the model; Ren et al. fine-tune retrieval and generation using legal training data that includes synthetic examples. Both approaches may improve task-specific output, but their success still depends on retrieving relevant evidence and checking whether the generated text is supported. The available studies do not establish that either approach inherently preserves statutory hierarchy or guarantees correct citations.

Legalyze places more emphasis on the index. By attaching document-location metadata to retrieved passages and combining keyword and semantic retrieval, it aims to make evidence easier to locate and verify. This can be useful where large-scale model fine-tuning is impractical, but it does not make the system independent of data quality: parsing errors, incomplete corpora, and weak retrieval can still affect the answer. Its design should therefore be assessed through retrieval metrics, citation-level evaluation, and end-to-end legal correctness, not through architecture alone.

None of these approaches is universally best. They address different questions: how to draft from retrieved precedent, how to adapt models to legal tasks, and how to preserve the structure of source documents during retrieval. A promising research direction is to combine them: a well-adapted generator working over a structure-aware index, followed by explicit verification that each material claim is supported by an authoritative source.

The practical direction is therefore not to choose between model adaptation and structure-aware retrieval. A system may use a fine-tuned retriever or generator when the data and resources justify it, while keeping the legal corpus external, versioned, and traceable. Retrieval, generation, and verification should remain distinct stages so that errors can be located, measured, and corrected.

8 Open problems and future directions

The review suggests five connected research directions. They are not isolated engineering improvements: each supports the same goal of making legal answers traceable, correct, and useful across jurisdictions.

Structure-preserving retrieval. Legal indexing should retain the hierarchy and context of each provision, not only the text of a chunk. Future work should compare flat chunking, document trees, breadcrumbs, and knowledge graphs using the same queries and corpus. Useful metrics include hierarchy accuracy, retrieval recall, ranking quality, and whether the returned context preserves exceptions and parent provisions.

Citation verification as a separate stage. A system should check that each cited authority exists, is current for the relevant date, belongs to the correct jurisdiction, and supports the claim attached to it. Evaluation should report citation existence, citation precision and recall, and claim-to-source entailment separately. These measures would help distinguish a plausible-looking answer from one whose legal basis can be audited.

Shared and realistic evaluation. Results from different papers are difficult to compare when they use different queries, corpora, and metrics. Legal RAG benchmarks should include retrieval-only and end-to-end tests, expert judgments, multiple question types, and error categories. They should also include statutes as well as case law, and report latency and cost alongside correctness. Benchmark coverage should be expanded across jurisdictions and languages rather than assuming that one country's legal tasks represent all legal work.

Beyond retrieval, legal AI also needs stronger reasoning analysis and governance. Dugac and Altwicker (2025) study the classification of legal interpretation methods, while Prince Tritto and Ponce (2025) review causal AI for legal language processing. These are related but distinct directions: neither should be treated as a substitute for source retrieval or citation verification. Kim et al. (2025) also highlight adoption challenges that involve trust, governance, and human oversight. An auditable pipeline can support oversight by making the retrieval, generation, and verification steps visible, but human review remains important for high-stakes use.

The evidence points to a practical principle: trustworthy legal AI should make each important step inspectable. Retrieval should show which passages were selected, generation should connect claims to evidence, and verification should test those connections. Governance and human oversight are easier to implement when the system's operations are separated and logged rather than hidden inside one opaque response.

These research directions reinforce one another. Preserving legal structure makes sources easier to locate; explicit provenance supports citation verification; and shared evaluation shows whether these benefits generalize across languages and jurisdictions. The next generation of legal AI should therefore be designed as a sequence of checkable steps (retrieve, generate, and verify) with clear failure measures for each stage.

9 Conclusion

Over more than three decades, legal AI has moved from explicit representations of cases and rules to learned legal language models, shared benchmarks, and retrieval-augmented systems. Each shift has expanded what computers can do with legal text, but none has re-

Table 3: Three design philosophies for modern legal AI, compared across the dimensions that matter for a legal tool. Buffa et al. and Ren et al. share a model-and-data outlook (adapt the generation or the weights); Hamza et al. adapt the index to preserve structure and provenance.

Dimension	**Buffa et al. (2025)**	**Ren et al. (2026)**	**Hamza et al. (2026)**
What is adapted	Generation prompt and retrieved context	Retriever and generator weights	Index and retrieval process
Core idea	RAG for legal document drafting	Synthetic-data generation, ranking, and fine-tuning	Structure-aware hierarchical indexing (SAHI)
Handling of structure	Retrieved text used as context; hierarchy is not the main contribution	Learned representations; explicit hierarchy preservation is not the main contribution	Breadcrumb metadata preserves each chunk's location
Retrieval	Semantic retrieval for drafting context	Fine-tuned dense retriever plus generator	BM25 and dense retrieval fused with RRF
Training-data need	No task-specific fine-tuning required by the described approach	Requires legal/synthetic training examples and training compute	No generator fine-tuning required; relies on reliable parsing and indexing
Provenance	Depends on retrieved material and implementation details	RAG provides context, but fine-tuning itself is not citation verification	Explicit source-location metadata accompanies retrieved passages
Reliability question	Does the draft faithfully use the retrieved context?	Do fine-tuning and retrieved evidence improve answer quality consistently?	Are parsing, ranking, and source metadata accurate?
Potential advantage	Practical document-drafting workflow	Task adaptation of retrieval and generation	Traceable, hierarchy-aware retrieval
Main risk / limitation	Draft requires review against authoritative sources	Training cost; synthetic-data quality; no guarantee against unsupported claims	Parser errors, corpus coverage, and retrieval errors can propagate

moved the need to justify an answer with authoritative evidence. Early case-based systems made reasoning visible but were difficult to scale. Legal language models processed text more flexibly, while benchmarks made their capabilities easier to compare. RAG brought external sources into the generation process, and reliability studies showed why retrieval alone is not enough.

The comparison of Buffa et al. (2025), Ren et al. (2026), and Hamza et al. (2026) illustrates three complementary priorities: improving drafting with retrieved context, adapting models to legal tasks, and preserving document structure during retrieval. The literature does not support a universal winner because these systems address different tasks and have not been compared under a common protocol. It does, however, point toward a common design goal: legal answers should be fluent, supported by relevant authority, and easy to verify.

The main research opportunities are therefore connected: structure-preserving retrieval, claim-level citation verification, shared evaluation, multilingual support, and broader jurisdictional coverage. A strong legal AI system must not only find and generate text; it must preserve where that text came from and show why it supports the answer. Future research should test these properties using transparent datasets, expert evaluation, and reproducible comparisons.

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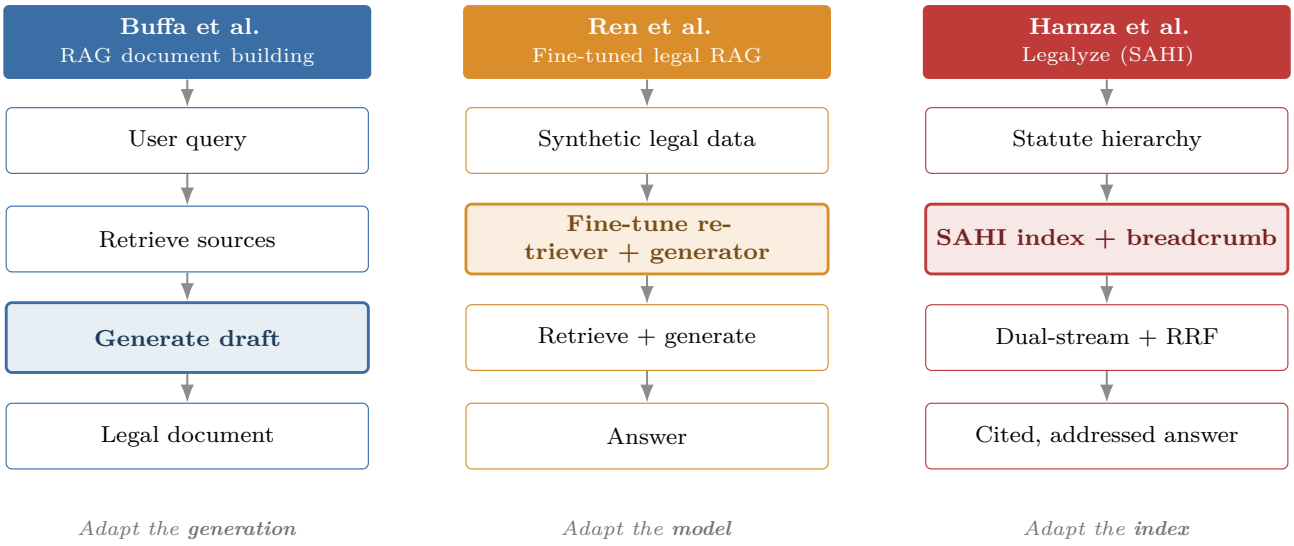


Figure 2: Three ways to build a modern legal system, drawn as parallel pipelines. Buffa et al. (2025) adapt the *generation* side: retrieve supporting text and let a model assemble a draft. Ren et al. (2026) go further and adapt the *model* itself, fine-tuning both the retriever and the generator on synthetic legal data. Hamza et al. (2026) instead adapt the *index*: they preserve the statute's hierarchy with a breadcrumb per chunk, retrieve through two streams fused by rank, and return each passage with its exact address. The highlighted box in each column marks that system's defining step. The three answer different questions, and a mature system is likely to combine them.

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